

**Lower
Premolars**

Dentition

Buccal aspect	Lower 1 st premolar	Lower 2 nd premolar
N.B	Has a large buccal cusp and a small nonfunctional lingual cusp which sometimes resemble the cingulum found on upper canine. 	The crown is wider than that of lower 4 but shorter due to wide neck. 
GEOL	Trapezoid with the smallest side cervically	
Outlines	Mesial: concave Distal: concave Occlusal outline: MCS and DCS are concave Mesial cusp slope is shorter than Distal cusp slope. Therefore the cusp tip is deviated mesially. Cervical line: is curved rootwise.	The same as lower 4 but:- -the buccal cusp of lower 5 is shorter than buccal cusp of lower 4. (due to more occlusal positioned of mesial and distal contact areas. -
Contact areas	Mesial CA: at the junction of middle and occlusal thirds Distal CA: more cervically	MCA: at the occlusal 1/3 DCA: at the junction between the middle and occlusal thirds.
Elevations	Buccal ridge : extends from the occlusal to cervical in the middle of the crown. Cervical Ridge : in cervical 1/3	The buccal ridge is less developed.
Depressions	Two shallow dev. grooves mesial and distal to buccal ridge	
Root	Is cone shape with tapered distally curved apex.	

Lingual aspect	Lower 1 st premolar	Lower 2 nd premolar
	Lingual convergence 	Less lingual convergence (it means that lingual surface is wider and longer than that of lower 4). 
	The lingual cusp is short and reach 2/3 the length of the buccal cusp.	The lower 5 has one or two lingual cusps , which is equal or even higher than the buccal cusp. If there are 2 lingual cusps, the mesiolingual cusp is larger than the distolingual cusp .
Landmarks	Mesiolingual developmental groove , originates from the pit at the bottom of the mesial triangular fossa crosses to the mesial surface near the mesiolingual line angle and fades in the middle third.	If there are 2 lingual cusps, the mesiolingual cusp is larger than the distolingual cusp , they will be separated by lingual developmental groove .
Root	Is similar to buccal aspect but is narrower.	

Mesial aspect	Lower 1 st premolar	Lower 2 nd premolar
GEOL	Rhomboid with narrow occlusal outline.	
Outlines	Buccal outline :convex with the maximum convexity at the cervical 1/3 Lingual outlines : convex with the maximum convexity at the middle third Cervical outline: slightly curves occlusaly. Occlusal outline: The lingual cusp is short and reach 2/3 the length of the buccal cusp. The crown is tilted lingually, due to severe lingual inclination. So the buccal cusp tip is at one line with the root apex. And the lingual cusp tip is in line with the lingual border of the root.  Mesial	Same as lower 4 but :- -the buccal cusp tip is less pointed . - there is less lingual inclination of the crown than that of lower 4. -the crown and root is wider buccolingually. - the lingual cusp may be longer than the buccal cusp. - the maximum convexity in the lingual outline may be in the occlusal third. (مهمة) -In case of 3 cusp types, if the crown is seen from mesial aspect, you can see only 2 cusps , the buccal and the mesiolingual , because the distolingual cusp is shorter than the mesiolingual cusp.  Mesial
Landmark	Mesial marginal ridge , severely slope from buccal to lingual .	Mesial marginal ridge , is not sloping.
N.B	Buccal outline is longer than lingual outline.	The root is longer and wider with blunt end.
Root	Wide and has deep groove	

Distal aspect	Lower 1 st premolar	Lower 2 nd premolar
	The same as the mesial aspect except : -the distal marginal ridge is perpendicular on the long axis of the tooth (no sloping). The groove on the root is deeper  Distal	The same as mesial aspect , but in case of 3 cusp types , there are 2 cusps on lingual , the mesiolingual cusp and the distolingual cusp. And the distolingual cusp is shorter than the mesiolingual cusp , so if the crown is seen from the distal aspect you will see all the 3 cusps  Distal

Occlusal aspect	Lower 1 st premolar	Lower 2 nd premolar	
GEOL	Diamond	2 Cusp type Oval	3 Cusp type Square
Dimensions	Thickness > Width	Thickness > Width	Thickness = Width
Lingual convergence	Severe lingual convergence	Less severe lingual convergence	No lingual convergence
Elevations	The buccal triangular ridge is larger than the lingual one. The two ridges are meeting in the transverse ridge . The MMR slopes from buccal to lingual. The DMR is straight.		Buccal cusp > mesiolingual cusp > distolingual cusp
Depressions	Mesial triangular fossa and distal triangular fossa, with 2 developmental pits so appears as snake eye appearance. The mesiolingual developmental groove extends from the mesial pit to the mesial surface. The central developmental groove crosses the transverse ridge.  Occlusal	The Central developmental groove is U or H shape pattern.  	The central developmental groove is Y shape There are 3 fossae :- 2 minor fossae : mesial and distal triangular fossae 1 minor fossa : central fossa 

Pulp cavities of lower premolars:

Mesiodistally:

- The pulp chamber is pointed occlusally with one pulp horn buccally and one pulp horn lingually .
- In case of 3 cusp type lower 5 , there are 3 pulp horns under each cusp.

	Begainig of calcification	Crown completion	Eruption	Root completion
4	1.5 – 1.75 years	3-4 years before eruption date	10 years	3-4 years after eruption date
4	1.75 - 2 years		10 years	
5	2 – 2.25 years		11 -12 years	
5	2.25 – 2.5 years		11 – 12 years	